# Cognitive Modeling

## Complete Lecture Notes

Week 1: What Is a Cognitive Model?

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# Chapter 1

# Fundamentals of Cognitive Modeling

## 1.1 Week 1: What Is a Cognitive Model?

> Chapter 1: Fundamentals of cognitive modeling · Lecture 1 h · Lab 2 h This week's running model: a simple random walk model of two-choice decisions.

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### 1.1.1 Key terms

Important terms of this lesson, with their Persian equivalents (repeated even if they appeared in earlier weeks).

Table 1.1: Key terms

| English | فارسی | توضیح کوتاه |
|---|---|---|
| cognitive model | مدل شناختی | توصیف دقیق و ریاضی یا محاسباتی از یک فرایند ذهنی |
| random walk | گام تصادفی | جمع شدن گام‌به‌گام شواهد تا رسیدن به یک مرز |
| verbal theory | نظریه‌ی کلامی | نظریه‌ای که فقط با کلمات بیان شده و پیش‌بینی دقیق نمی‌دهد |
| Marr's levels | سطوح تحلیل Marr | سطح محاسباتی، الگوریتمی و تحلیل |
| parameter | پارامتر | کمیت قابل تنظیم مدل که بین افراد یا شرایط فرق می‌کند |
| noise | نویز | تغییرات تصادفی در شواهد یا در داده |
| simulation | شبیه‌سازی | تولید داده‌ی مصنوعی با اجرای مدل |

| English | فارسی | توضیح کوتاه |
|---|---|---|
| drift rate | نرخ رانش | سرعت میانگین جمع شدن شواهد؛ نشانه‌ی کیفیت شواهد |
| boundary separation | فاصله‌ی مرزها | مقدار شواهد لازم برای پاسخ؛ نشانه‌ی احتیاط |
| evidence | شواهد | اطلاعاتی که برای تصمیم جمع می‌شود |
| accuracy | دقت | نسبت پاسخ‌های درست |
| response time (RT) | زمان واکنش | فاصله‌ی ارائه‌ی محرک تا پاسخ |
| formal model / formalize | فرمال کردن / مدل فرمال | نوشتن نظریه به شکل معادله یا الگوریتم |
| fit / fitting | برازش | تنظیم پارامترها تا مدل با داده جور شود |
| forward / inverse problem | معکوس / مسئله‌ی مستقیم | شبیه‌سازی (پارامتر ← داده) در برابر برآورد (داده ← پارامتر) |
| non-decision time | زمان غیرتصمیمی | زمان دیدن محرک و اجرای حرکت پاسخ |
| speed–accuracy trade-off | بده‌بستان سرعت و دقت | سریع‌تر پاسخ دادن به قیمت خطای بیشتر |
| trial | کوشش (trial) | یک بار انجام تکلیف توسط شرکت‌کننده |
| temperature | دما (temperature) | پارامتر نویز در انتخاب token؛ $\beta = 1/T$ |
| random seed | seed تصادفی | عدد شروع مولد اعداد تصادفی برای تکرارپذیری |
| quantile | چندک (quantile) | مقداری که درصد مشخصی از داده زیر آن است |
| large language model (LLM) | مدل بزرگ زبانی | شبکه‌ی عصبی‌ای که کلمه‌ی بعدی متن را پیش‌بینی می‌کند |
| token | token (واحد متن) | کلمه یا بخشی از کلمه که مدل زبانی پیش‌بینی می‌کند |

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### 1.1.2 Learning outcomes

By the end of this week, students will be able to:

1. Define a cognitive model and distinguish it from a verbal theory and from a purely statistical model of the data.
2. Classify models by purpose (descriptive, process, normative) and by family (symbolic, connectionist, stochastic/dynamical, Bayesian), and place them on Marr's three levels.
3. Explain how free parameters, a deterministic core and a noise model together turn a model into a *data-generating* machine.
4. Interpret how changes in the model's parameters (drift, boundary) map onto psychological constructs (evidence quality, caution).

### 1.1.3 Motivation & big picture

Consider a familiar verbal theory of simple decisions:

> *"When people decide, they gather evidence until they have enough, then they respond."*

It sounds complete, but try to get a prediction out of it:

- If we tell participants to *be accurate* rather than *be fast*, how much slower should they get? Twice as slow? 50 ms slower?
- Are errors faster or slower than correct responses, or equally fast?
- Should the response time (RT) distribution be symmetric or skewed?
- If a stimulus becomes harder to see, should RTs change in the same way as under an accuracy instruction?

The verbal theory cannot answer any of these. Each answer depends on decisions the theory never made: What counts as "evidence"? Does it arrive continuously or in chunks? What does "enough" mean? Is the threshold the same for both responses?

As soon as we write the theory down as a model, for example a random walk that steps up or down until it reaches a boundary, every one of those questions gets a definite answer. Some answers will turn out to be *wrong*, and that is useful: a model that can be wrong is a model we can learn from. This is the central message of the course:

> Formal models force us to make our assumptions explicit, produce precise predictions, and turn "Is this theory right?" into "How well, and why, does this model account for the data?"

Where is this used? Models like the random walk of this week are not only classroom examples. They are used every day in research and practice:

- Clinical psychology and psychiatry. Patients with depression, ADHD or anxiety are often "slower" in simple tasks. Decision models show *why*: lower evidence quality, more caution, or slower motor responses. These three explanations call for different interventions.
- Ageing research. Older adults respond more slowly. Fitting decision models showed that much of this slowing comes from more caution and slower motor execution, not from worse information processing.
- Neuroscience. Neurons in some brain areas increase their firing rate until a threshold before a decision, much like a random walk reaching a boundary. Model parameters are used to link behavior and brain activity.

- Applied settings. Eyewitness decisions, medical image reading and quality control all involve deciding under noisy evidence. The same model separates "how good is the evidence" from "how careful is the decision maker".
- Artificial intelligence. Ideas from cognitive models (evidence accumulation, learning from errors, probabilistic inference) are used in AI systems, and AI models are now tested with the methods of this course (see the AI & LLMs section).

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### 1.1.4 Lecture notes

In plain words A *theory* says something in words ("people collect evidence until they have enough"). A *model* says the same thing so precisely that a computer can run it. Once it can run, the model makes exact predictions: how often people will be correct, how long they will take, and how both change when we change the task. This week's model is the simplest possible decision model. Imagine a token on a number line. Each moment it moves one step up (a bit of evidence for the correct answer) or one step down (a bit of evidence for the wrong answer). When it reaches the top line, the person answers correctly; when it reaches the bottom line, they make an error. Only three numbers control everything: how strongly the steps lean upward (drift, the quality of the evidence), how far apart the two lines are (boundary separation, how careful the person is), and where the token starts (starting point, bias).

Psychology words in plain language

- Participant (subject): a person who takes part in an experiment.
- Stimulus (plural stimuli): whatever is shown or played to the participant, e.g. a picture, a word or a tone.
- Task: what the participant must do, e.g. "press left if the dots move left, right if they move right". A two-choice task has exactly two possible responses.
- Trial: one repetition of the task: one stimulus, one response. An experiment usually has hundreds of trials per participant.
- Response time (RT, reaction time): the time from stimulus onset to the key press, usually in milliseconds (ms).
- Accuracy: the proportion of correct responses.
- Condition / manipulation: a change the experimenter makes on purpose (e.g. "be fast!" vs. "be accurate!" instructions, or easy vs. hard stimuli), to see how behavior changes.
- Construct: a psychological quantity we cannot observe directly, e.g. *caution*, *attention* or *memory strength*. A good model parameter *measures* a construct.
- Behavioral data: what we can record: choices and RTs (not brain signals).

Suggested timing (60 min): motivation 10 · what and why 10 · kinds of models + Marr 15 · models as data-generating machines + pipeline 10 · first model (random walk) 12 · notation and course logistics 3.

## 1. What is a cognitive model?

A cognitive model is an explicit, usually mathematical or computational, description of the processes or representations that are assumed to produce behavior. It has three ingredients:

Table 1.2: 1. What is a cognitive model?

| Ingredient | Question it answers | Random walk example |
|---|---|---|
| Assumptions / structure | What mechanism do we posit? | Evidence accumulates in discrete steps between two boundaries |
| Parameters | Which quantities are free to vary across people or conditions? | Drift $v$, boundary separation $a$, start point $z$ |
| Mapping to data | How does the mechanism produce observable responses? | The boundary reached gives the choice; the number of steps gives the RT |

Three related terms are easy to mix up:

- A theory is a set of verbal (often informal) claims about the mind.
- A model is one particular formal instantiation of a theory. One theory can be formalized in many different ways.
- A statistical model (a t-test, a regression) is also a model, but of the *data*. Its parameters (means, slopes) usually describe the data rather than a cognitive process. In a cognitive model the parameters are meant to measure latent psychological quantities, such as memory decay, caution or attention.

George Box's often-quoted remark that "all models are wrong, but some are useful" sums up the attitude of this course. We never ask whether a model is *true*. We ask whether it is *useful*: does it capture the important regularities, generalize to new data, and have interpretable parameters?

## 2. Why bother with formal models?

1. Precision. Writing the model down exposes hidden assumptions (as in the motivation example).
2. Quantitative predictions. Models predict *how much*, not just *whether*.
3. Deriving surprising consequences. Our intuitions about complex systems are poor. For example, averaging individual exponential learning curves produces a curve that looks like a *power* law (Week 5).
4. Measurement. Fitted parameters are measurements of latent constructs. $d'$ in signal detection theory (Week 2) measures sensitivity separately from response bias, which raw accuracy cannot do.

5. Comparing theories. Two verbal theories that "both explain the data" can be pitted against each other once they are formalized (Weeks 6–7, 10).
6. Bridge to AI. The same formalisms (probabilistic models, learning rules, neural networks) are used to build artificial agents (Week 15).

### 3. Kinds of cognitive models

By purpose:

Table 1.3: 3. Kinds of cognitive models

| Type | Question | Example | Course weeks |
|---|---|---|---|
| Descriptive | *What* is the shape of the data? | Power law of practice; exponential forgetting curve | 2–7 |
| Process (mechanistic) | *How* does the mind produce the behavior? | Random walk / drift-diffusion; exemplar model (GCM) | 1, 14 |
| Normative (rational) | What *should* an ideal agent do? | Bayesian ideal observer for cue combination | 15 |

The boundaries are fuzzy: an SDT model is partly descriptive and partly a process model.

By family:

Table 1.4: 3. Kinds of cognitive models

| Family | Core idea | Example |
|---|---|---|
| Symbolic / rule-based | Cognition as manipulation of symbols and production rules | ACT-R production systems |
| Connectionist | Distributed representations, learning by weight change | Delta-rule networks, deep networks |
| Stochastic / dynamical | Noisy processes unfolding in time | Random walk, drift-diffusion, Rescorla–Wagner |
| Bayesian (probabilistic) | Cognition as probabilistic inference | Bayesian observer models |

> Watch out: "Bayesian modeling" in this course means two different things. (a) Bayesian models of cognition claim that the *mind* does Bayesian inference (Week 15). (b) Bayesian data analysis is how *we, the scientists,* estimate the parameters of any model, Bayesian or not (Weeks 9–13). Weeks 9–13 are mostly about (b).

### 4. Marr's three levels of analysis

David Marr proposed analyzing an information-processing system at three complementary levels:

Table 1.5: 4. Marr's three levels of analysis

| Level | Question | Decision example |
|---|---|---|
| Computational | What problem is being solved, and why? | Choose the correct alternative while balancing speed and accuracy |
| Algorithmic / representational | What representations and procedures solve it? | Accumulate noisy evidence until a boundary is reached |
| Implementational | How is it realized physically? | Firing rates of neurons ramping up to a threshold |

Normative models usually live at the computational level, process models at the algorithmic level, and computational neuroscience models at the implementational level. Most models in this course are algorithmic-level models.

### 5. A model is a data-generating machine

The single most important idea for the rest of the course: a model specifies a probability distribution over possible data.

$$\mathrm{y} \sim p(\mathrm{y} \mid \boldsymbol{\theta}, \mathcal{M})$$

Read this as: "under model $\mathcal{M}$ with parameters $\boldsymbol{\theta}$, the data y are generated from the distribution $p$." Many models split this into a deterministic core plus noise, for example $y_i = f(x_i; \boldsymbol{\theta}) + \varepsilon_i$ (Week 2). Others, like the random walk, are stochastic all the way through.

This gives two directions of reasoning:

Table 1.6: 5. A model is a data-generating machine

| Direction | Given | Find | Name | Course weeks |
|---|---|---|---|---|
| Forward | $\boldsymbol{\theta}$ | Possible data y | Simulation / prediction | 1–2 |

| Direction | Given | Find | Name | Course weeks |
|---|---|---|---|---|
| Inverse | Observed y | Plausible $\boldsymbol{\theta}$ | Parameter estimation / fitting | 3–5, 9–13 |

The inverse problem is harder. It is what we mean by "fitting a model to data". Because a model can always be run forwards, we can also test our fitting methods: simulate data from known parameters $\boldsymbol{\theta}^*$, fit, and check whether we get $\boldsymbol{\theta}^*$ back (*parameter recovery*, Week 7).

Reading the math $\quad \mathrm{y} \sim p(\mathrm{y} \mid \boldsymbol{\theta}, \mathcal{M})$ is read: "the data y are drawn from the probability distribution $p$, given the parameter values $\boldsymbol{\theta}$ and the model $\mathcal{M}$."

- y (bold) is a *list* of observations, e.g. the 200 responses of one participant.
- The symbol $\sim$ means "is randomly generated from". It reminds us that the same model with the same parameters gives *different* data each time, just as a person does not respond identically on every trial.
- The vertical bar $\mid$ means "given" or "assuming". Everything to the right of it is fixed: the model's equations ($\mathcal{M}$) and the numbers we plug into them ($\boldsymbol{\theta}$).
- $\boldsymbol{\theta}$ (theta, bold) collects all parameters, e.g. $\boldsymbol{\theta} = (v, a, z)$ for the random walk.

Example. For a coin with $P(\text{heads}) = 0.6$, the model is "each flip is heads with probability $\theta$", $\boldsymbol{\theta} = 0.6$, and y could be (H, T, H, H, T). Run it again and you get another sequence. *Forward* direction: choose $\theta$, simulate y. *Inverse* direction (fitting, Weeks 3–5): see y, find a good $\theta$.

### 6. The modeling pipeline (our roadmap)

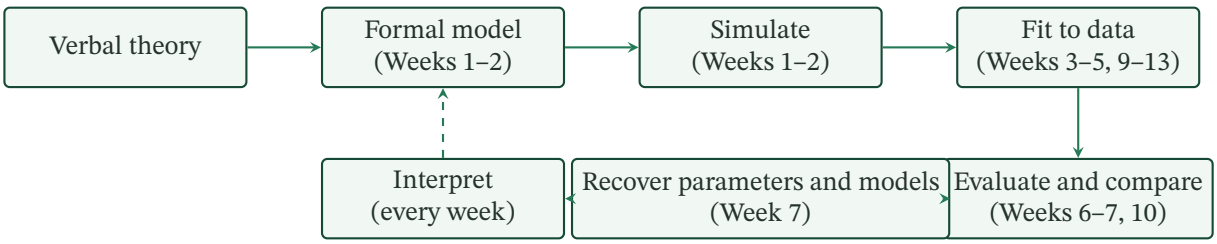



Figure 1.1: The cognitive modeling pipeline from Week 1

Every lab in this course runs some version of this loop. Today we focus on formalize $\rightarrow$ simulate $\rightarrow$ interpret, with a small preview of fit.

### 7. Our first model: a random walk to a boundary

Verbal $\rightarrow$ formal. Evidence is a single number $X$ that starts at $z$. At each time step it moves up by 1 (evidence for the correct response) or down by 1 (evidence for the error response). The walk stops when $X$ reaches the upper boundary $a$ (correct response) or the lower boundary $0$ (error).

$$X_0 = z, \qquad X_n = X_{n-1} + \delta_n, \qquad \delta_n = \begin{cases} +1 & \text{with probability } \frac{1+v}{2} \\ -1 & \text{with probability } \frac{1-v}{2} \end{cases}$$

- $v \in (-1, 1)$ is the drift, the average step $\mathbb{E}[\delta_n] = v$. It represents the quality of evidence: easy stimuli give a large $v$.
- $a$ is the boundary separation. It represents caution: a wider gap needs more evidence.
- $z$ is the start point ($0 < z < a$). With $z = a/2$ the decision maker is unbiased.
- The number of steps until absorption, $T$, maps onto RT through

$$\text{RT} = t_0 + T\,\Delta t,$$

  where $t_0$ is non-decision time (encoding plus motor execution) and $\Delta t$ is the duration of one step.

Analytic predictions. Let $r = \frac{1-v}{1+v}$. For $v \neq 0$, the probability of a correct response and the mean number of steps are:

$$P_c = \frac{1 - r^z}{1 - r^a}, \qquad \mathbb{E}[T] = \frac{a\,P_c - z}{v}.$$

These two equations already settle the questions from the motivation:

- Increasing $a$ raises $P_c$ and $\mathbb{E}[T]$: a speed–accuracy trade-off.
- Increasing $v$ raises $P_c$ but lowers $\mathbb{E}[T]$: easier stimuli are both faster and more accurate.

So the two manipulations leave different signatures in the data. That is what lets us *measure* caution and evidence quality separately.

> Advanced: where the formulas come from. $P_c$ is the classic *gambler's ruin* result. Let $h(x)$ be the probability of hitting $a$ before $0$ when starting from $x$. Conditioning on the first step gives $h(x) = \frac{1+v}{2}h(x+1) + \frac{1-v}{2}h(x-1)$ with $h(0) = 0$ and $h(a) = 1$. Solving this difference equation gives $h(x) = (1 - r^x)/(1 - r^a)$. For $\mathbb{E}[T]$, use *Wald's identity*: the final position satisfies $\mathbb{E}[X_T] = z + v\,\mathbb{E}[T]$. Since $X_T = a$ with probability $P_c$ and $X_T = 0$ otherwise, $\mathbb{E}[X_T] = aP_c$, and rearranging gives $\mathbb{E}[T] = (aP_c - z)/v$. As the step size and $\Delta t$ shrink together, the random walk becomes the drift-diffusion model (Week 14).

> Advanced: a property that turns out to be a weakness. With an unbiased start ($z = a/2$), the simple random walk predicts that correct and error responses have identical RT distributions. Empirically, errors are often *slower* than correct responses when accuracy is stressed, and sometimes *faster* under speed stress. The simple model cannot produce either pattern. Adding trial-to-trial variability in

drift or start point can (lab Exercise 5). Finding where a model fails is exactly how models improve.

Reading the math The update rule. $X_n = X_{n-1} + \delta_n$ says: *the evidence now = the evidence one step ago + one new step*. The step $\delta_n$ is $+1$ or $-1$. The probability of $+1$ is $\frac{1+v}{2}$. With $v = 0$ this is $0.5$ (a fair coin: no information). With $v = 0.2$ it is $0.6$, so the walk drifts upward on average. The average step is $(+1)(0.6) + (-1)(0.4) = 0.2 = v$, which is why $v$ is called the drift.

The accuracy formula. $P_c = \dfrac{1 - r^z}{1 - r^a}$ with $r = \dfrac{1 - v}{1 + v}$.

- $r$ is the ratio "probability of a down step / probability of an up step". With $v = 0.2$: $r = 0.4/0.6 = 2/3$. A small $r$ means the walk almost always goes up.
- $r^z$ and $r^a$ are small numbers when $z$ and $a$ are large, so $P_c$ approaches 1. That is the math behind "more caution (larger $a$) gives higher accuracy".
- You do not need to derive the formula (see the Advanced box). It is enough to plug in numbers and check it against a simulation, as in the lab.

The time formula. $\mathbb{E}[T] = \dfrac{aP_c - z}{v}$. $\mathbb{E}[\cdot]$ means "average over many trials". The numerator is how far the walk travels on average (it ends at $a$ with probability $P_c$, at $0$ otherwise, and started at $z$). Dividing by the average speed $v$ gives the average number of steps, just like *time = distance / speed*.

RT. RT $= t_0 + T\,\Delta t$: a fixed part $t_0$ (seeing the stimulus, pressing the key) plus the decision time, i.e. the number of steps times the duration of one step.

For an engineer You already know this model under another name.

- Sequential detection. The random walk is a discrete version of Wald's sequential probability ratio test (SPRT): accumulate the log-likelihood ratio of the observations, and decide as soon as it crosses an upper or a lower threshold. Under some assumptions the SPRT gives the smallest average number of observations for given error probabilities, which is one reason the brain might use this strategy.
- Drift ≈ signal-to-noise ratio. Each step is "signal + noise". The drift $v$ is the mean of a step and the randomness of the step is the noise, so $v$ plays the role of the SNR per sample. A cleaner stimulus is a higher SNR.
- Boundary ≈ decision threshold. Raising the threshold lowers the error probability but increases the detection delay: the classic trade-off between error rate and latency in sequential detection. In psychology this is called the speed–accuracy trade-off.
- Start point ≈ prior. Starting closer to one boundary is like using a non-zero initial log-likelihood ratio, i.e. a prior in favor of one hypothesis.
- Non-decision time ≈ fixed processing delay. Sensor/encoding latency plus actuator (motor) latency, added to every response.
- What is new for an engineer: in engineering we *design* the detector. Here we *observe* a human detector and try to infer its parameters (SNR, threshold) from its decisions and decision times. That inverse problem is the subject of the whole course.

### 8. Course-wide notation

This table is used in all weeks. When a later week introduces a symbol, it uses the meaning given here.

Data and design

Table 1.7: 8. Course-wide notation

| Symbol | Meaning | First used |
|---|---|---|
| $y_i$, $\mathrm{y} = (y_1, \ldots, y_N)$ | Observation $i$; vector of all observations | Wk 1 |
| $N$ | Number of observations (trials) | Wk 1 |
| $x_i$ | Experimental condition / independent variable of observation $i$ | Wk 2 |
| $j = 1, \ldots, J$ | Participant index; number of participants | Wk 5 |

Models and parameters

Table 1.8: 8. Course-wide notation

| Symbol | Meaning | First used |
|---|---|---|
| $\mathcal{M}$, $\mathcal{M}_1, \mathcal{M}_2$ | A model; competing models | Wk 1 |
| $\boldsymbol{\theta}$ | Vector of free parameters | Wk 1 |
| $k$ | Number of free parameters | Wk 1 |
| $\boldsymbol{\theta}^*$ | True (data-generating) parameter values in a simulation | Wk 1 |
| $\hat{\boldsymbol{\theta}}$ | Estimate of $\boldsymbol{\theta}$ | Wk 3 |
| $f(x; \boldsymbol{\theta})$ | Deterministic part of a model's prediction | Wk 2 |
| $\hat{y}_i$ | Model prediction for observation $i$ | Wk 3 |
| $\varepsilon_i$ | Noise (error) term | Wk 2 |

Probability

Table 1.9: 8. Course-wide notation

| Symbol | Meaning |
|---|---|
| $P(\cdot)$ | Probability of an event |
| $p(\cdot)$ | Probability mass or density function |
| $y \sim \mathcal{D}$ | "$y$ is distributed as $\mathcal{D}$" |
| $\mathcal{N}(\mu, \sigma)$ | Normal distribution with mean $\mu$ and standard deviation $\sigma$ (as in and ) |
| Bernoulli$(\theta)$, Binomial$(n, \theta)$ | Bernoulli and binomial distributions with success probability $\theta$ |
| Beta$(a_0, b_0)$, Uniform$(l, u)$ | Beta and uniform distributions |
| $\Phi$, $\Phi^{-1}$ | Standard normal CDF and its inverse |
| $\mathbb{E}[\cdot]$, Var$[\cdot]$ | Expectation and variance |

Fitting, evaluation and Bayesian inference

Table 1.10: 8. Course-wide notation

| Symbol | Meaning | First used |
|---|---|---|
| SSE, MSE, RMSD | Sum of squared errors, mean squared error, root mean squared deviation | Wk 3 |
| $L(\boldsymbol{\theta}) = p(\text{y} \mid \boldsymbol{\theta}, \mathcal{M})$ | Likelihood | Wk 4 |
| $\ell(\boldsymbol{\theta}) = \ln L(\boldsymbol{\theta})$; NLL $= -\ell$ | Log-likelihood; negative log-likelihood | Wk 4 |
| $K$ | Number of cross-validation folds (not the same as $k$!) | Wk 6 |
| AIC, BIC | Akaike and Bayesian information criteria | Wk 7 |
| $p(\boldsymbol{\theta})$, $p(\boldsymbol{\theta} \mid \text{y})$ | Prior; posterior | Wk 9 |
| $p(\text{y} \mid \mathcal{M})$ | Marginal likelihood (evidence) | Wk 10 |
| $\text{BF}_{12}$ | Bayes factor for $\mathcal{M}_1$ over $\mathcal{M}_2$ | Wk 10 |
| $\boldsymbol{\theta}^{(s)}$, $s = 1, \ldots, S$ | $s$-th MCMC draw; number of draws | Wk 11 |
| $C$ | Number of MCMC chains | Wk 12 |
| $\hat{R}$, ESS | Potential scale reduction factor (R-hat); effective sample size | Wk 12 |

Model-specific symbols

Table 1.11: 8. Course-wide notation

| Model | Symbols | First used |
|---|---|---|
| Random walk / drift-diffusion | $v$ drift; $a$ boundary separation; $z$ start point; $t_0$ non-decision time; $\Delta t$ step duration; $T$ number of steps | Wk 1, 14 |
| Signal detection theory | $H$ hit rate; $F$ false-alarm rate; $d'$ sensitivity; $c$ criterion | Wk 2 |
| Forgetting curves | $t$ retention interval; $A$ initial level; $\lambda$ exponential decay rate; $d$ power-law decay exponent | Wk 2 |
| GCM (categorization) | $w_m$ attention weights; $c$ sensitivity | Wk 14 |
| Reinforcement learning | $\alpha$ learning rate; $\beta$ inverse temperature; $Q_t(o)$ value of option $o$ on trial $t$ | Wk 15 |

Overloaded symbols (kept because the literature uses them): $c$ is the SDT criterion except in Week 14, where it is the GCM sensitivity. $t$ is the retention interval in forgetting models and the trial index in RL. $\alpha$ is only used as the RL learning rate. We never use $\alpha$ for a significance level.

For the instructor

- Start with the questions, not the model. Read the verbal theory aloud and ask students to answer the four questions of the Motivation section. They will disagree. That disagreement is the reason formal models exist.
- Act out the random walk. Draw a vertical line on the board with a top line ("correct") and a bottom line ("error"). Let a student roll a die: 1–4 means up, 5–6 means down ($v \approx 0.33$). Count steps until a line is reached. Do this three times and you have three trials with different RTs. Then move the lines further apart and repeat.
- Keep the math light in this week. Students only need to *use* the formulas for $P_c$ and $\mathbb{E}[T]$, not derive them. Show that the numbers match the simulation in the lab: that is the main message (simulation and analysis agree).
- Typical student questions.
  - *"Is evidence really discrete steps?"* No. It is a simplification; Week 14 shows the continuous version (drift-diffusion), which gives similar predictions.
  - *"Why not just compare mean RTs with a t-test?"* A t-test says *that* groups differ; the model says *how* (caution vs. evidence quality).
  - *"Which model is the true one?"* None. All models are simplifications; we ask which one is useful (Box's quote).

- Time management. Sections 3–4 (model types, Marr) can be shortened to 10 minutes if needed. Do not skip Section 7: everything in the lab depends on it.

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### 1.1.5 Worked example

Setup. An unbiased decision maker with drift $v = 0.2$, boundary separation $a = 6$ and start point $z = 3$. Non-decision time $t_0 = 300$ ms; each step lasts $\Delta t = 20$ ms.

Step 1: step probabilities. $P(\delta = +1) = (1 + 0.2)/2 = 0.6$ and $P(\delta = -1) = 0.4$.

Step 2: simulate one trial by hand. Draw uniform random numbers $u_n$. Step up if $u_n < 0.6$, otherwise step down.

Table 1.12: Worked example

| Step $n$ | $u_n$ | $\delta_n$ | $X_n$ |
|---|---|---|---|
| 0 | — | — | 3 |
| 1 | 0.31 | +1 | 4 |
| 2 | 0.72 | −1 | 3 |
| 3 | 0.55 | +1 | 4 |
| 4 | 0.12 | +1 | 5 |
| 5 | 0.48 | +1 | 6 = a $\rightarrow$ stop |

The walk hits the upper boundary: a correct response after $T = 5$ steps, so RT $= 300 + 5 \times 20 = 400$ ms.

Step 3: analytic accuracy. $r = (1 - 0.2)/(1 + 0.2) = 2/3$, so $r^3 = 8/27 \approx 0.296$ and $r^6 = 64/729 \approx 0.088$.

$$P_c = \frac{1 - 8/27}{1 - 64/729} = \frac{1}{1 + 8/27} = \frac{27}{35} \approx 0.771.$$

(When $z = a/2$, the formula simplifies to $P_c = 1/(1 + r^{a/2})$.)

Step 4: analytic mean RT.

$$\mathbb{E}[T] = \frac{6 \times 0.771 - 3}{0.2} = \frac{57/35}{0.2} \approx 8.14 \text{ steps}, \qquad \mathbb{E}[\text{RT}] = 300 + 8.14 \times 20 \approx 462.9 \text{ ms}.$$

Step 5: two manipulations, two signatures.

Table 1.13: Worked example

| Condition | $v$ | $a$ ($z = a/2$) | $P_c$ | $\mathbb{E}[T]$ | $\mathbb{E}$[RT] (ms) |
|---|---|---|---|---|---|
| Baseline | 0.2 | 6 | 0.771 | 8.14 | 462.9 |
| "Be accurate!" (more caution) | 0.2 | 10 | 0.884 | 19.18 | 683.6 |
| Easier stimulus (better evidence) | 0.3 | 6 | 0.865 | 7.30 | 446.0 |

Interpretation. Accuracy instructions buy about 11 percentage points of accuracy at a cost of roughly 220 ms. Easier stimuli give a *similar* accuracy gain while making responses slightly *faster*. A researcher who looked only at accuracy could not tell the two situations apart. The model, by using accuracy *and* RT together, attributes them to different psychological causes (caution vs. evidence quality).

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### 1.1.6 Lab (2 hours)

#### Goals

1. Formulate and evaluate the random walk model, simulate an experiment, and verify the simulation against the analytic formulas.
2. Use the model to interpret a speed–accuracy trade-off.
3. Get a first taste of fitting: recover the hidden parameters of a "mystery participant", evaluate the fit, and interpret the result.

Timing: Part A 35 min · Part B 15 min · Part C 30 min · Part D 15 min · Part E 25 min. Exercises not finished in the lab are finished at home.

#### Part C: Formulate, simulate and verify the random walk (30 min)

First, look at what the model does: plot a few evidence trajectories.

Now simulate a whole experiment and compare the simulation with the analytic predictions. Always check a simulation against something you know; this habit will save you many times this term.

What to notice: - The simulation matches the analytic values to within sampling noise. - The RT distributions are right-skewed (a long tail of slow responses), which is the typical shape of empirical RT distributions. - Correct and error RTs are almost identical (see the Advanced box in Lecture §7).

### Part D: Interpret, the speed–accuracy trade-off (15 min)

Moving along one curve (increasing $a$) trades speed for accuracy. Moving *between* curves (increasing $v$) improves accuracy at the same or lower RT. The table reproduces the worked example: e.g., $v = 0.2, a = 10$ gives accuracy 0.884 and mean RT 683.6 ms.

Discuss with your neighbor: an experimenter finds that a drug makes participants slower *and* more accurate. Which parameter would you suspect changed? What if the drug made them slower and *less* accurate?

### Part E: A first fit, evaluation and interpretation (25 min)

Don't peek: try to recover them.

We assume $t_0 = 300$ ms and $\Delta t = 20$ ms are known and $z = a/2$. Only $v$ and $a$ are free ($k = 2$).

Fitting by grid search. For every combination of $v$ and $a$ on a grid, compute the predicted accuracy and mean RT and measure how far they are from the observed values. Each difference is expressed in standard-error units so that accuracy and RT are on a comparable scale. Week 3 develops this idea properly.

Evaluating the fit. We fitted only two summary statistics (accuracy and mean RT). A stronger test is whether the model also reproduces something we did *not* fit: the shape of the RT distribution. We compare RT quantiles of correct responses in the data with those of a large simulation at the best-fitting parameters.

The hidden values were $v^* = 0.16$ and $a^* = 10$. The grid search recovered $a$ exactly and $v$ to within one grid step (0.15). The small miss is not a bug: with 1000 trials, the observed accuracy and mean RT differ from their true values by sampling noise, so the best-fitting parameters do too. (Week 5 shows how to put an uncertainty interval around estimates like these.) RTs are multiples of 20 ms, which is why many quantiles match exactly. The model reproduces the shape of the distribution well, with only a slight underestimate of the slow tail (0.9 quantile).

In psychological terms, what do the recovered $v$ and $a$ say about this participant? 2. The fit used only two numbers from the data. Why is the quantile comparison a more convincing check than the discrepancy value itself? 3. If this participant were re-tested under "be as fast as possible" instructions, which parameter should change and which should stay the same, according to the model?

### Exercises (increasing difficulty)

Exercise 1 $\star$ (sampling noise). Simulate the baseline model ($v = 0.2$, $a = 6$) with $N = 20, 100$ and $1000$ trials, five different seeds each. Make a table of the observed accuracies. Compare their spread with the standard error $\sqrt{P_c(1 - P_c)/N}$. *Expected:* with $N = 20$ accuracies can easily range from about 0.6 to 0.95. The spread shrinks roughly with $1/\sqrt{N}$.

Exercise 2 $\star$ (two conditions). Simulate a "speed" condition ($a = 4$) and an "accuracy" condition ($a = 10$) with the same $v = 0.2$, 1000 trials each. Report accuracy and mean RT per condition and plot both RT distributions. Which aspect of the RT distribution changes most: its minimum, its mode, or its tail?

Exercise 3 ⋆⋆ (response bias). Set $a = 8$ and $z = 5$ (start closer to the correct boundary). Compute $P_c$ analytically and by simulation. Then compare the mean RT of correct and error responses. Explain the result in terms of the distance each boundary is from the start point. What would $z = 3$ do?

Exercise 4 ⋆⋆ (checking a model property). For $z = a/2$, verify by simulation (20 000 trials) that correct and error RT *quantiles* are equal to within sampling noise. Empirical data often show slower errors in difficult, accuracy-stressed conditions. Is the simple random walk falsified by such data? What would you change?

$v_i \sim \mathcal{N}(v, \eta)$ Use $v = 0.2$, $\eta = 0.2$, $a = 8$. Show that errors are now slower than correct responses (expected with 20 000 trials: accuracy ≈ 0.75, about 12 steps for correct vs. 14 steps for error responses), and explain why. *Hint:* which trials are most likely to end in an error, and how long do those trials take?

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### 1.1.7 Connection to AI & LLMs

Marr's levels for a chatbot. A large language model (LLM) can be described at all three of Marr's levels, and the levels can come apart just as they do for people. *Computational:* the model is trained to predict the next word (token) in a text. *Algorithmic:* a transformer network computes a score (a "logit") for every possible next token and turns the scores into probabilities. *Implementational:* matrix multiplications on GPUs. A claim like "the LLM understands the question" is usually made at the computational level, while the evidence (its outputs) is produced by the algorithm. Keep the levels apart when you read such claims.

A familiar decision rule. The final step of an LLM is a choice rule with a noise parameter, just like the cognitive models in this course. The probability of token $i$ is a softmax with temperature $T$: $P(i) = e^{z_i/T} / \sum_j e^{z_j/T}$. A low $T$ gives near-deterministic choices and a high $T$ gives more random ones. This is the same rule as the softmax in Week 15, with $\beta = 1/T$.

- ⋆ Is the LLM's temperature closer to the random walk's *noise* or to its *boundary* (Week 1, Part C)? Explain. - ⋆⋆ Discussion: someone says "the chatbot gave a different answer when I asked again, so it does not know the answer." Use this box to explain what else could be going on.

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### 1.1.8 Common mistakes & misconceptions

1. "A model that fits well is true." Good fit is necessary but not sufficient. A flexible model can fit almost anything, including noise (Week 6). Fit says little until you know what the model *could not* have fitted.
2. "More parameters make a better model." More parameters make a model *more flexible*, which is not the same as better. We will penalize flexibility explicitly (AIC/BIC, cross-validation, Bayes factors).

3. Confusing simulation with fitting. Simulation goes from parameters to data; fitting goes from data to parameters. Students often "fit" by tweaking parameters until a plot looks nice. That is a legitimate *exploration* but not an estimation method.
4. Treating parameters as directly observable. $v$ and $a$ are *inferred* quantities whose values depend on the model's assumptions (e.g., that $t_0$ is known). Change the model and the "same" parameter may change meaning.
5. Not verifying simulations. A bug in a simulator produces wrong conclusions that *look* plausible. Compare against an analytic result, a limiting case (e.g., $v \to 0$ should give $P_c \to z/a$), or a published value.
6. Mixing up the two meanings of "Bayesian". Bayesian models of cognition (a claim about the mind) vs. Bayesian data analysis (a method for scientists).
7. Treating Marr's levels as competing. They are complementary descriptions of the same system, not rival theories.

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### 1.1.9 Check your understanding

1. In one sentence each, define a *theory*, a *cognitive model* and a *statistical model*. Answer A theory is a set of (usually verbal) claims about mental processes. A cognitive model is a formal, explicit instantiation of such claims that can generate predictions, with parameters that represent psychological quantities. A statistical model describes the structure of the data (e.g., group means, slopes) without committing to a cognitive mechanism.

2. Classify each as descriptive, process or normative: (a) RT $= A\,N^{-b}$ as a function of practice trials $N$; (b) the random walk model; (c) an ideal observer that combines visual and haptic cues weighted by their reliabilities. Answer (a) Descriptive: it summarizes the shape of the data without saying why. (b) Process: it specifies a mechanism that produces choices and RTs. (c) Normative: it specifies optimal behavior given the available information.

3. At which of Marr's levels does the random walk model mainly live? Give an example of a claim about the same decision at the implementational level. Answer The algorithmic/representational level: it describes a procedure (accumulate evidence to a boundary). An implementational claim would be, e.g., "the accumulated evidence is carried by the firing rate of neurons in a specific brain area, and the response is triggered when that rate reaches a threshold."

4. What is the difference between the forward and the inverse problem? Which one did we solve in Part E of the lab? Answer Forward: given parameters, generate (predict) data, i.e., simulation. Inverse: given data, infer the parameters, i.e., fitting. Part E solved an inverse problem (by grid search), but it relied on solving the forward problem many times, once per grid point.

5. Using the formulas, what happens to $P_c$ when $v = 0$? (Hint: think about the limit, or about the symmetry of the walk.) Answer With no drift the walk is symmetric, and the probability of hitting the upper boundary equals $z/a$ (the limit of the formula as $v \to 0$). With an unbiased start, $z = a/2$, accuracy is 0.5, i.e., chance. That is sensible: with no evidence the decision maker is guessing.

6. Two groups differ in accuracy (0.80 vs. 0.90). Why can't we tell from accuracy alone whether they differ in caution or in evidence quality? What extra data resolve this? Answer Both a larger

boundary $a$ and a larger drift $v$ raise accuracy, so accuracy alone confounds them. RT resolves it: higher caution makes responses slower, while better evidence makes them faster (worked example, Step 5). Fitting the model to accuracy and RT jointly separates the two.

7. Why did we simulate data from *known* parameters before trusting the fitting procedure? Answer Only with known true values can we check whether the method returns the right answer (parameter recovery). If a method cannot recover parameters from data generated by the model itself, its estimates from real data are not interpretable. This is developed fully in Week 7.

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### 1.1.10 Assignment

#### A0: Verbal theory audit

A0 does not carry points, but it is required: an incomplete A0 must be revised before the Week 2 lab check-off can be awarded.

Part 2: Verbal theory audit (max. 1 page). Choose one verbal theory from your earlier courses, for example "levels of processing", "attention as a spotlight", "practice makes perfect", "dual-process theories of reasoning" or "interference causes forgetting".

1. State the theory in 1–2 sentences.
2. List at least five decisions the theory leaves unspecified that a formal model would have to make (e.g., functional form, what is noisy, what the free parameters are, how internal states map onto the measured behavior).
3. For two of these decisions, propose a concrete formal choice (an equation or an algorithmic step).
4. Describe one prediction that would differ depending on which choice is made, and an experiment that could tell them apart.

Feedback criteria (pass / revise):

Table 1.14: A0: Verbal theory audit

| Criterion | Pass | Revise |
|---|---|---|
| Underspecification | $\geq 5$ distinct, genuinely unspecified decisions | Fewer than 5, or restatements of the theory |
| Formalization | Two concrete, unambiguous formal choices | Vague ("use a function") |
| Diverging prediction | A specific prediction plus an experiment that distinguishes the choices | No prediction, or a prediction that does not depend on the choice |

The best audits will be discussed briefly at the start of the Week 2 lecture.

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### 1.1.11 Further reading

- Farrell & Lewandowsky (2018), chapters 1–2. Introduction to computational modeling and the move from verbal theories to formal models. *(Main reading for this week.)*
- Verguts (2022), introductory chapter [verify]. A short, accessible motivation for modeling cognitive processes.
- Anderson (2014), introductory chapter [verify]. Places cognitive modeling alongside computational neuroscience, useful context for Marr's implementational level.
- *Optional classic (outside the course list):* Marr, D. (1982). *Vision.* W. H. Freeman. The original source of the three levels of analysis.

Assignment A1 will be released.